Vehicle Motion Control System

High-Level Design (HLD)

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Revision: B

Status: Approved for pilot evaluation

Classification: Confidential - internal engineering reference

This document describes the software architecture of the vehicle motion control domain: components, ports, interfaces, signals, functional flows and basic software integration.

1 Revision history

Revision	Date	Author	Description
A	2026-04-10	R. Iyer	Initial architecture baseline
B	2026-08-22	R. Iyer	Traction control extension, steering rename

2 Scope and context

2.1 Scope

The vehicle motion control domain covers longitudinal and lateral motion commands: vehicle speed processing, brake intervention, steering angle handling and gateway exchange with the powertrain and body domains.

Out of scope: actuator calibration values, production software details and hardware pin mapping. Those are covered by separate interface specifications.

2.2 Traction control context

Revision B adds an advisory traction control assist path that reuses the existing chassis gateway and brake arbitration components.

2.3 System context

The domain runs on BodyEcu and ChassisEcu. Gateway exchange with the powertrain domain is performed by PowertrainGateway over the chassis CAN bus.

3 Software architecture

3.1 Component breakdown

Component	Type	Description
VehicleSpeedManager	ApplicationSwComponentType	Filters and plausibilises wheel sp
BrakeController	ApplicationSwComponentType	Brake torque arbitration and distr
SteeringAnglePreprocessor	ApplicationSwComponentType	Steering angle preprocessing and f
ChassisSignalService	ServiceSwComponentType	Chassis signal gateway service
PowertrainGateway	EcuAbstractionSwComponentType	Powertrain domain gateway
BodyControlAdapter	ApplicationSwComponentType	Body electronics adaptation
DiagnosticsFrontend	ApplicationSwComponentType	Diagnostic services front end
VehicleMotionComposition	CompositionSwComponentType	Composition of the domain componen
TractionControlAssist	ApplicationSwComponentType	Advisory traction control assist

3.2 Ports and interfaces

Port	Direction	Interface	Component
VehicleSpeedOut	provided	VehicleSpeedI	VehicleSpeedManager
WheelSpeedIn	required	WheelSpeedI	VehicleSpeedManager
BrakePressureOut	provided	BrakePressureI	BrakeController
SteerAngleOut	provided	SteeringAngleI	SteeringAnglePreprocessor
ChassisGatewayIn	required	ChassisSignalSetI	ChassisSignalService
GatewayToPowertrain	provided	PowertrainExchangeI	PowertrainGateway
LampCommandOut	provided	LampCommandI	BodyControlAdapter
DiagRequestIn	required	DiagnosticServiceI	DiagnosticsFrontend
TractionRequestOut	provided	TractionRequestI	TractionControlAssist

VehicleSpeedOut port is provided by VehicleSpeedManager and consumed by PowertrainGateway. SteeringAngleOut is required by BrakeController for lateral stability arbitration.

PowertrainGateway requires VehicleSpeedOut at 10 ms to update the gateway frame, and requires GearPosition for the shift intervention path.

The Chassis_SignalService gateway keeps its legacy spelling in the powertrain gateway mapping table.

3.3 Signal inventory

Signal	Interface	Sender	Receiver	Bits
VehicleSpeed	VehicleSpeedI	VehicleSpeedManager	PowertrainGateway	16
WheelSpeedFL	WheelSpeedI	ChassisSignalService	VehicleSpeedManager	16
WheelSpeedFR	WheelSpeedI	ChassisSignalService	VehicleSpeedManager	16
BrakePressure	BrakePressureI	BrakeController	DiagnosticsFrontend	12
SteeringAngle	SteeringAngleI	SteeringAnglePreprocess	BrakeController	12
GearPosition	GearPositionI	PowertrainGateway	BodyControlAdapter	8
LampCommand	LampCommandI	BodyControlAdapter	DiagnosticsFrontend	8
TractionRequest	TractionRequestI	TractionControlAssist	BrakeController	8

4 Functional flows

4.1 Vehicle speed flow

WheelSpeedFL and WheelSpeedFR arrive via ChassisSignalService and are consumed by VehicleSpeedManager. VehicleSpeedManager uses WheelSpeedSensorAdapter for plausibility checks before publishing VehicleSpeed through the VehicleSpeedI interface.

PowertrainGateway receives VehicleSpeed and republishes it to the powertrain domain. The same value is requested by DiagnosticsFrontend for read data by identifier services.

4.2 Brake intervention flow

BrakeController requires BrakePressureI from the brake pedal path and receives SteeringAngle from SteeringAngleReader. When the stability threshold is exceeded, BrakeController requests torque reduction from PowertrainGateway through PowertrainExchangeI.

4.3 Traction control flow

TractionControlAssist publishes TractionRequest over TractionRequestI. BrakeController consumes the request and applies wheel individual braking. TorqueVectoringBlock uses TractionControlUnit to coordinate the advisory torque limit with the powertrain domain.

BodyControlAdapter raises AmberWarningSignal while the assist function is active, and clears the lamp once the request is released.

5 Timing and runnables

TimingEvent VehicleSpeedSampleEvent triggers RunnableEntity UpdateVehicleSpeedRunnable every 10 ms. TimingEvent BrakeCheckEvent triggers EvaluateBrakePressureRunnable every 20 ms.

Both runnables are scheduled by MotionTask on ChassisEcu. The composition runnable ComposeVehicleMotionRunnable executes after the sensor runnables and before the gateway runnable PublishGatewayValuesRunnable.

6 Basic software integration

Signal routing uses Com and PduR. Incoming frames are handled by CanIf on both ECUs. NvM stores the calibration block for the speed plausibility window. Rte provides the port API used by all application components.

Mode management is performed by BswM and EcuM. Watchdog supervision for MotionTask is performed by WdgM. The chassis gateway maps IPdu to NPdu inside PowertrainGateway before transmission.

The diagnostic path routes UDS requests through Dcm and Dem. DiagnosticsFrontend consumes DiagnosticServiceI and forwards routine results to the tester.

7 Diagnostics and configuration

DTCs for speed plausibility, brake pressure range and gateway timeout are defined in the diagnostics specification. The HLD only defines the reporting path: component to Dcm to CanIf.

Configuration parameters are grouped into the VMC configuration container and written through NvM during shutdown.

8 Known limitations

TBD.